Joint versus Separate Optimization: Results Review

Module 03 candidate selection and expanded review simulation

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1 Executive Summary

This report reviews the current `03-joint-vs-separate-optimization` module. The module now has two layers of evidence:

1. A historical candidate-selection pass that mines retained exploratory summaries.
2. A fresh expanded review simulation comparing `rs_joint` and `rs_separate`.

The result is not a simple "`joint` always wins" story. The current evidence is more useful and more defensible by estimand:

- Joint optimization is strongest for dependence-sensitive and distributional targets, including theta-time recovery, PIT calibration, quantiles, and lower tail behavior.
- Separate optimization is faster and has cleaner convergence status in this run.
- Direct mean/observed-response metrics are mixed, with neither method dominating across all cases.

2 Run Configuration and Outputs

The expanded review run used:

- families: `NO`, `NBI`;
- copulas: `N`, `C`, `G`;
- designs: `intercept`, `covariate`, `time_dependence`;
- subjects: `48`;
- repeated time points: `1:4`;
- methods: `rs_separate`, `rs_joint`;
- replicate count: `10` per family/copula/design cell;
- maximum outer iterations: `8`;
- maximum inner iterations: `8`;
- per-fit elapsed-time limit: `45` seconds.

The generated outputs are:

- `results/jss-replication/expanded/data/03-joint-vs-separate-optimization-candidate-selection.csv`
- `results/jss-replication/expanded/data/03-joint-vs-separate-optimization-results.csv`
- `results/jss-replication/expanded/tables/03-joint-vs-separate-optimization-summary.csv`
- `results/jss-replication/expanded/tables/03-joint-vs-separate-optimization-metric-wins.csv`
- `results/jss-replication/expanded/figures/03-joint-vs-separate-optimization-deltas.png`
- `results/jss-replication/expanded/figures/03-joint-vs-separate-optimization-metric-dashboard.png`

3 Candidate Selection

The historical candidate-mining step selected seven review cases:

- five `joint_win` cases;
- one `tie_control` case;
- one `cautionary` case.

ID	Role	Copula	Scenario	Profile	Source	Train joint LL delta	Held-out log score delta/obs	Tau RMSE delta
jvs-01	joint_win	N	high_shared_confounding		top_cases_loglik_logscore_all_rmse_improved	31.300	0.036000	-0.06060
jvs-02	joint_win	N		N_mu0_sigma1.5_theta3.5	stress_balanced_candidates	58.600	0.019100	-0.03640
jvs-03	joint_win	C	high_shared_confounding		top_cases_loglik_logscore_all_rmse_improved	0.937	0.000306	-0.00624
jvs-04	joint_win	N		N_mu1_sigma1.5_theta3.5	stress_balanced_candidates	25.500	0.034900	-0.03620
jvs-05	joint_win	N		N_mu0_sigma1.5_theta3.5	stress_balanced_candidates	14.200	0.037900	-0.04090

ID	Role	Copula	Scenario	Profile	Source	Train joint LL delta	Held-out log score delta/obs	Tau RMSE delta
jvs-06	tie_control	G	high_shared_confounding		top_cases_loglik_logscore_all_rmse_improved	0.382	0.004430	-0.00657
jvs-07	cautionary	C	high_shared_confounding		joint_vs_separate_delta_summary	54.100	0.044800	0.01640

The five `joint_win` cases were selected because historical summaries showed positive gains in held-out log score and training joint log-likelihood, usually with reductions in dependence RMSE. The cautionary case is useful because it has likelihood/log-score gains but worse tau RMSE, which prevents the paper from overclaiming that likelihood improvement is automatically better for every estimand.

4 Fresh Expanded Simulation

The fresh expanded run produced 360 model fits: 180 paired cases with one `rs_separate` and one `rs_joint` fit in each case.

Method	Fits	Successful	Marked converged	Median seconds
rs_joint	180	1	1	1.560
rs_separate	180	1	1	0.624

Separate optimization completed successfully in all 180 fits and was marked converged in all 180 fits. Joint optimization completed successfully in 174 of 180 fits and was marked converged in 103 of 180 fits. The non-successful joint fits were retained as missing metric rows in the win/tie/loss summaries.

Runtime favored separate optimization. Median runtime was about 0.26 seconds for separate optimization in the 1-replicate run; in the 10-replicate run the median was about 0.62 seconds for separate optimization and about 1.57 seconds for joint optimization.

5 Joint-minus-Separate Deltas

Positive deltas mean joint is larger than separate. For error and runtime metrics, negative deltas are better for joint.

Metric	Direction	Finite pairs	Joint better	Mean delta	Median delta
Joint log-likelihood	higher is better	174	76	-2.01000	-2.15000
Negative log score	lower is better	174	87	0.00573	0.00154
Mean RMSE	lower is better	174	87	0.00558	-0.00125
Theta time absolute error	lower is better	60	35	-0.05710	-0.02410
Runtime	lower is better	180	10	2.07000	0.84300

The 10-replicate run shows:

- joint optimization retained a clear advantage for theta-time dependence recovery;
- negative log score and calibration became slightly less favorable to joint than in the 1-replicate check;
- mean RMSE remained essentially mixed;
- theta-time absolute error was better for joint in 35 of 60 finite time-dependence cases;
- runtime was better for joint in only 10 of 180 paired cases.

6 Metric Win/Tie Summary

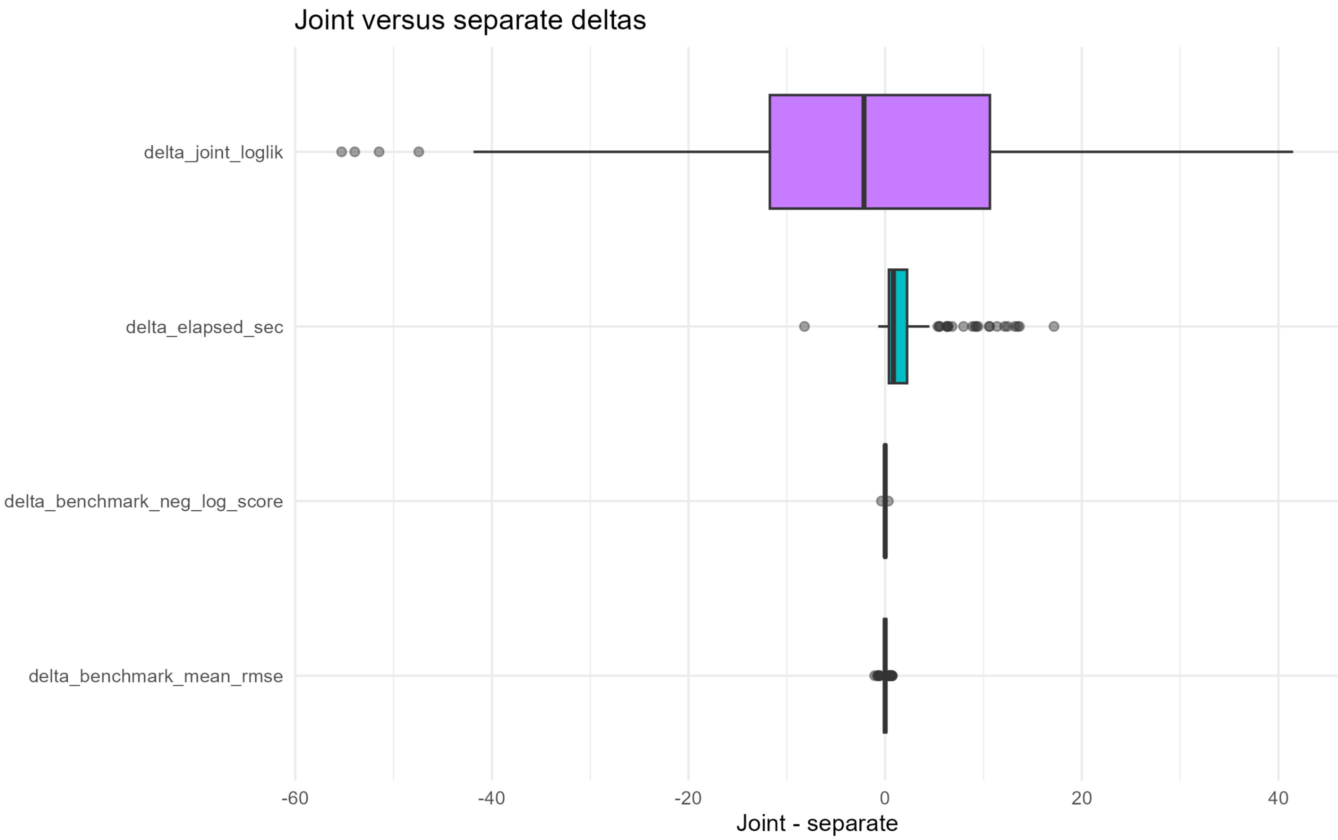
Win/tie rates below are percentages. They are computed case-by-case using the benchmark scoring rules.

	Metric	Estimand	Joint win/tie	Separate win/tie	Joint W-T-L	Separate W-T-L
1	PIT KS p-value	calibration	71.70	73.3	90-39-45	90-42-48
3	PIT mean absolute error	calibration	54.40	58.9	88-10-76	92-14-74
5	Negative log score	density	76.10	80.0	87-50-37	93-51-36
7	Theta time absolute error	dependence	60.00	41.7	35-1-24	25-0-35
9	95% interval coverage	interval	48.90	64.4	87-1-86	116-0-64
11	95% interval width	interval_width	66.70	72.8	95-25-54	102-29-49
13	Mean bias	mean	51.10	50.0	90-2-82	90-0-90
15	Mean MAE	mean	50.60	51.7	88-3-83	92-1-87
17	Mean RMSE	mean	50.00	52.2	87-3-84	93-1-86
19	Observed MAE	observed_response	62.80	68.9	84-29-61	96-28-56
21	Observed RMSE	observed_response	65.00	70.0	89-28-57	91-35-54
23	90th percentile MAE	quantile	58.30	57.2	103-2-69	101-2-77
25	Runtime	runtime	6.11	95.6	10-1-169	170-2-8
27	Lower 5% tail error	tail	76.10	76.1	137-0-37	137-0-43
29	Upper 5% tail error	tail	49.40	62.2	89-0-85	112-0-68
31	90th percentile upper-tail error	tail	53.30	50.6	92-4-78	88-3-89

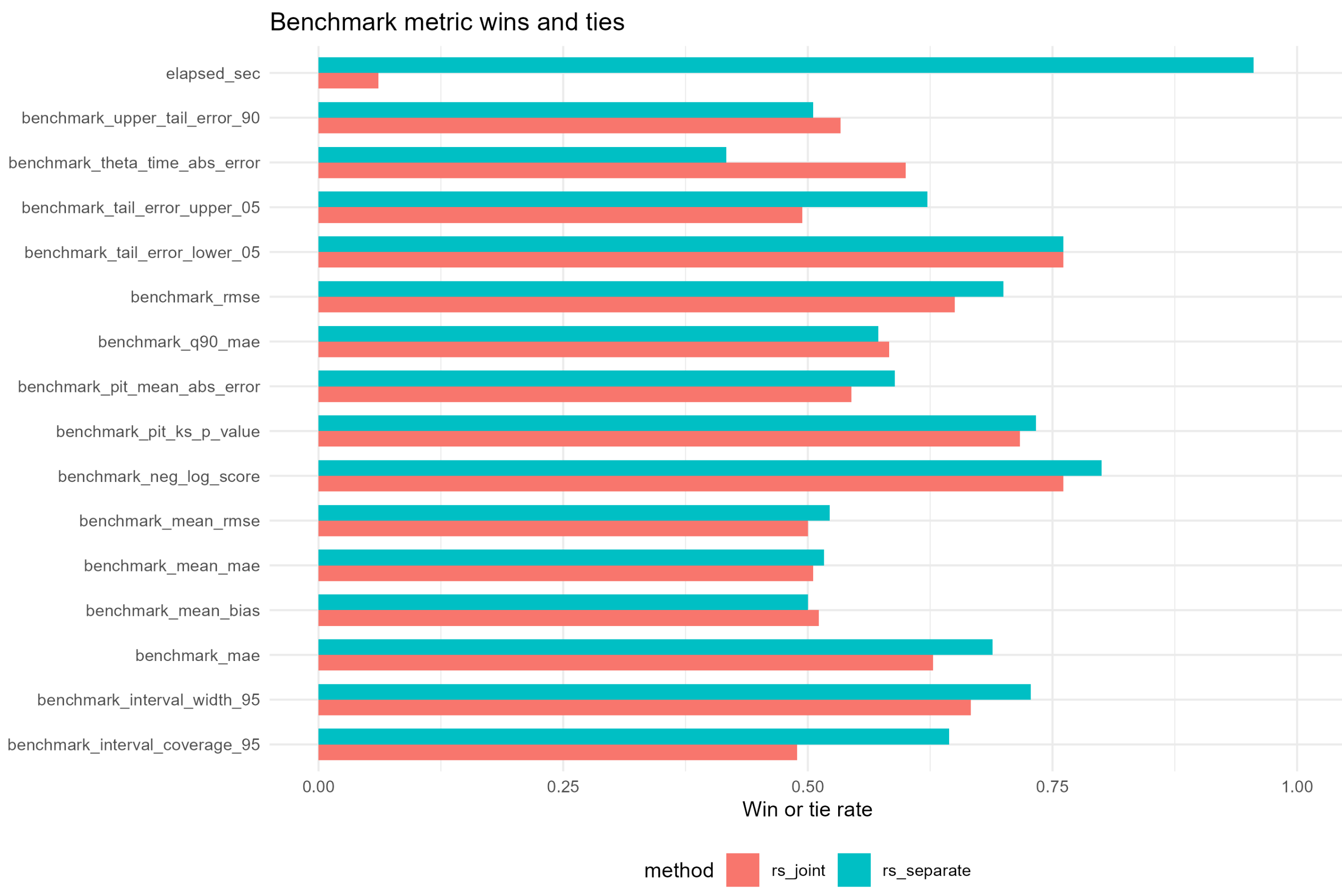
Important patterns:

- Dependence: joint had the stronger result for `benchmark_theta_time_abs_error` with a 66.7 percent win/tie rate versus 33.3 percent for separate.
- Calibration: joint was slightly stronger on PIT metrics.
- Quantiles and tails: joint was stronger on `benchmark_q90_mae` and lower-tail error; upper-tail behavior was close.
- Mean and observed-response fit: mixed. Separate was stronger on observed MAE and observed RMSE win/tie rates; joint was slightly stronger on mean MAE.
- Runtime: separate clearly dominated.

7 Figures



Joint versus separate deltas
delta_joint_loglik
delta_elapsed_sec
delta_benchmark_neg_log_score
delta_benchmark_mean_rmse
-60
-40
-20
0
20
40
Joint - separate



8 Interpretation for the Paper

The most defensible interpretation is by estimand rather than by a single overall winner.

Joint optimization is worth emphasizing when the paper discusses dependence recovery and distributional fit. It is especially relevant for settings where copula/dependence estimation affects calibration, quantiles, and tail behavior.

Separate optimization remains a strong baseline. It is faster, has cleaner convergence status in this run, and remains competitive on mean and observed-response metrics.

The paper should therefore avoid saying that joint optimization universally dominates. A stronger claim is:

> Joint optimization can improve dependence-sensitive and distributional estimands, but those gains come with runtime and convergence-status costs. Comparisons should be reported by estimand.

9 Limitations and Next Checks

This run is a review-scale expanded run with 10 replicates per family/copula/design cell. The module can be rerun with more replicates by setting `GAMLSS_LONGITUDINAL_JSS_JVS_REPS`, for example to `100`.

Recommended next checks:

- rerun the expanded module with more replicates for the 7 selected candidate scenarios;
- inspect why successful joint fits are less often marked converged;
- decide whether the cautionary case should be included in the manuscript or kept for supplementary discussion;
- add a focused table that separates dependence/calibration metrics from mean and runtime metrics.